

Data Analysis with R and Python

Semestral Examination

22 MAY 2026, 2:00PM – 3:30PM

Bachelor of Statistical Data

Science

TOTAL POINTS: 25

PART A INSTRUCTIONS

There are a total of 27 points on this paper. The maximum you can score is 25.

Write your answers on the question paper. Write only the answers to the questions asked. Do not write any R code or any other explanation unless instructed.

This is a practical examination; you may use laptops or desktops with R and necessary packages pre-installed, as per instructions given earlier. However, you are not allowed to access the internet during the examination.

In the following questions, a “confidence interval” refers to the standard confidence interval for the population mean μ based on a random sample from a $N(\mu, \sigma^2)$ distribution, based on the t distribution of

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}.$$

LCL and UCL refer to lower and upper confidence limits respectively.

Preliminary steps: We will use the NHANES dataset contained in the NHANES R package, and in particular the variables Age, Gender, Race1, and Height. Obtain a subset of the data that includes only the above four columns, and only the rows that satisfy all of the following conditions:

- Race1 is White, Black, or Mexican.
- Gender is male.
- The value of Height is not missing.

Further subset this dataset to form the following data frames:

- **subdataA**: consisting of all participants aged between 25 and 35 (inclusive)
- **subdataB**: consisting of all participants aged between 65 and 75 (inclusive)

1. The number of observations in **subdataA** (age 25–35) is [1]

2. The number of observations in **subdataB** (age 65–75) is [1]

3. The average Height ($\bar{X}$) in the three subgroups defined by **Race1** in the two datasets, rounded to 2 places after the decimal, are [3]

	White	Black	Mexican
subdataA			
subdataB			

4. The sample standard deviation (s) of Height in the three subgroups defined by **Race1** in the two datasets, rounded to 2 places after the decimal, are [3]

	White	Black	Mexican
subdataA			
subdataB			

5. The 90% confidence interval for the population mean heights of the three subgroups defined by **Race1** in **subdataA**, rounded to 2 places after the decimal, are [3]

	White	Black	Mexican
LCL			
UCL			

6. The 90% confidence interval for the population mean heights of the three subgroups defined by **Race1** in **subdataB**, rounded to 2 places after the decimal, are [3]

	White	Black	Mexican
LCL			
UCL			

7. Given two independent random samples

$$X_1, X_2, \dots, X_m \sim N(\mu_1, \sigma_1^2)$$

and

$$Y_1, Y_2, \dots, Y_n \sim N(\mu_2, \sigma_2^2),$$

we want to test the null hypothesis

$$H_0 : \mu_1 = \mu_2.$$

Consider the following test procedure: First, construct the 90% confidence intervals for μ_1 and μ_2 based on the respective samples. Reject H_0 if the two confidence intervals are disjoint, and accept it otherwise.

- (a) Using a simulation study, estimate the size of this test for the following combinations. Use 10^5 replications of the experiment for each combination. Report your estimate rounded to 3 places after the decimal. You may assume $\mu_1 = \mu_2 = 0$. [6]

- $m = 100, n = 200, \sigma_1 = 1, \sigma_2 = 1$

- $m = 100, n = 200, \sigma_1 = 1, \sigma_2 = 2$

- $m = 15, n = 20, \sigma_1 = 1, \sigma_2 = 1$

- $m = 15, n = 20, \sigma_1 = 1, \sigma_2 = 2$

- (b) Based on the above results and other checks you may perform, do you think the size of the test depends on [4]

- the sample sizes m and n ? YES ☐ NO ☐

- the standard deviations σ_1 and σ_2 ? YES ☐ NO ☐

- (c) Briefly comment on whether you think this is a reasonable test procedure, keeping in mind that in practice m and n are known, but σ_1 and σ_2 are unknown. [3]